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Inverse Modeling and Convex Optimization for Audience Vote Estimation and Voting Rule Evaluation

Abstract

In *Dancing with the Stars*, competition outcomes are jointly determined by judges' scores and audience votes, yet actual audience voting data are not publicly available, making it difficult to quantitatively evaluate the fairness of the voting mechanism. To address this incomplete-information setting, this paper formulates audience vote estimation as an inverse problem with ordering constraints, and reconstructs weekly audience voting proportions based on observed judges' scores and historical elimination paths.

Specifically, we develop a constrained optimization model in which audience vote shares are treated as decision variables and elimination consistency is enforced through a system of inequality constraints. To mitigate the non-uniqueness of inverse problems and to obtain behaviorally plausible solutions, the model incorporates a maximum entropy objective to characterize voting uncertainty, together with a temporal smoothing regularization term that penalizes abrupt changes in audience preferences across consecutive weeks. The resulting formulation is shown to be a convex optimization problem, guaranteeing the existence of a globally optimal solution and ensuring numerical stability and reproducibility.

Using the estimated audience votes as inputs, we conduct a counterfactual comparison of alternative voting aggregation rules, including percentage-based aggregation, rank-based aggregation, and two-stage mechanisms with judges' intervention. Quantitative metrics such as elimination consistency, elimination margins, and vote entropy are introduced to evaluate how different rules affect elimination outcomes and fairness across competition stages. In addition, stepwise regression models with fixed effects are employed to analyze the heterogeneous effects of professional dancers and celebrity characteristics on judges' evaluations and audience preferences.

Finally, based on the technical findings, we propose an improved elimination mechanism with dynamic weighting, which enhances fairness and stability while preserving audience engagement. A representative season case study is used to demonstrate the effectiveness of the proposed rule. Overall, this study illustrates how inverse modeling and optimization techniques can be applied to the analysis and design of voting mechanisms in competitive elimination systems.

Keywords: inverse modeling; constrained optimization; maximum entropy; temporal smoothing; voting rule comparison; fairness analysis

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1 Introduction

1.1 Problem Background

Dancing with the Stars (DWTS) is a reality television show centered on celebrity dance competitions. The competition results are jointly determined by professional judges' scores and audience votes. Judges' scores mainly reflect the technical quality of the dances, whereas audience voting is often influenced by subjective factors such as a celebrity's popularity and public image. As a result, significant discrepancies frequently arise between the two. Across different seasons, the show has repeatedly adjusted the method used to combine judges' scores and audience votes, and certain outcomes have sparked controversies regarding fairness. However, because actual audience voting data are not publicly available, the rationality of the voting mechanism is difficult to evaluate quantitatively. Based on the available judges' scores, contestant information, and elimination results, constructing a mathematical model to estimate audience votes and analyze the effects of different voting rules is of great importance for understanding the fairness of the competition mechanism and for optimizing future competition formats.

1.2 Clarifications and Restatements

Given the background information and available data conditions of this problem, we are required to complete the following tasks:

Task 1: In the absence of publicly available audience voting data, develop a mathematical model to estimate the audience votes received by each contestant based on the weekly judges' scores and the actual elimination outcomes across seasons. Evaluate the rationality and consistency of the estimated results.

Task 2: Using the estimated audience voting results obtained in Task 1, conduct a comparative analysis of the rank-based combination method and the proportional combination method used by the show. Examine the differences in competition outcomes under different voting combination rules across seasons, and analyze how each method affects the relative influence of judges' scores versus audience votes.

Task 3: Based on the dataset incorporating the estimated audience votes, construct a model to analyze the impact of professional dancers and celebrity personal characteristics (such as age, industry background, etc.) on competition performance. Compare the differences in their effects on judges' scores and audience voting.

Task 4: Integrating the findings from the previous analyses, propose a new mechanism for combining judges' scores and audience votes. Justify the proposed scheme in terms of fairness, stability, and/or entertainment value, and provide recommendations for optimizing the show's competition format.

1.3 Our Work

To address the tasks of this problem, we develop a unified analytical framework to study audience voting behavior and the fairness of voting mechanisms.

First, in the absence of publicly available audience vote data, we formulate the problem as an inverse modeling task. By enforcing consistency with observed elimination outcomes, we estimate weekly audience voting proportions and characterize their uncertainty.

Second, using the estimated audience votes as inputs, we compare different voting aggregation rules, including the percentage-based rule, the rank-based rule, and mechanisms involving judges' intervention, and examine their impacts on elimination outcomes.

Next, we analyze the effects of professional dancers and celebrity characteristics on judges' scores and audience votes using regression-based methods.

Finally, based on the above analyses, we propose an improved elimination mechanism and evaluate its performance through a representative season case study.

2 Preparation for Modeling

2.1 Model Assumptions

To ensure the solvability of the model and the reasonableness of the analysis, the following assumptions are made:

1. Within the same season, the total number of audience votes in each week is constant, and audience support for contestants can be represented by vote proportions.

2. The judges' scores released by the show objectively reflect the technical quality of each contestant's performance in that week and are assumed to be accurate.

3. The observed elimination outcomes strictly follow the officially announced score aggregation and elimination rules, without considering editing effects or temporary rule changes.

4. Under the percentage-based aggregation mechanism, a contestant's combined score is given by the linear sum of the judges' score share and the audience vote share.

5. Audience preferences evolve smoothly across consecutive weeks, without abrupt and unexplained changes.

6. In the absence of additional information, the distribution of audience votes tends to satisfy the maximum entropy principle, avoiding excessively concentrated voting patterns.

2.2 Notations

Table 1: Notation and Variable Definitions

Symbol	Type	Description
s	Index	Season index
t	Index	Week index within season s
i, j	Index	Contestant indices
$A_{s,t}$	Set	Set of contestants remaining in the competition at week t
$E_{s,t}$	Set	Set of contestants eliminated at week t
$k_{s,t}$	Parameter	Number of eliminated contestants in week t, $k_{s,t} = \lvert E_{s,t} \rvert$
$J_{s,i,t}$	Given parameter	Judges' score of contestant i in week t
$q_{s,i,t}$	Intermediate variable	Normalized judges' score proportion, $q_{s,i,t} = \frac{J_{s,i,t}}{\sum_{j \in A_{s,t}} J_{s,j,t}}$
$v_{s,i,t}$	Decision variable	Audience voting proportion of contestant i in week t, satisfying $v_{s,i,t} \geq 0$ and $\sum_{i \in A_{s,t}} v_{s,i,t} = 1$
$C_{s,i,t}$	State variable	Combined score of contestant i in week t, $C_{s,i,t} = q_{s,i,t} + v_{s,i,t}$
ε	Parameter	Small positive constant used to avoid ranking ties and enhance numerical stability
λ	Parameter	Temporal smoothing coefficient controlling week-to-week variation of audience votes
$H_{s,t}$	Objective term	Entropy of audience voting distribution in week t
$S_{s,t}$	Objective term	Temporal smoothness penalty between consecutive weeks

2.3 Data Preprocessing

The original data are recorded in a wide-table format, containing basic information on contestants from each season as well as weekly judges' scores. To meet the requirements of modeling, this study first converts the original wide-format data into a long-table structure indexed by season–week–contestant. The multiple judges' scores received by each contestant within the same week are then aggregated to obtain the contestant's weekly total judges' score and average score.

To address inconsistencies in the number of judges, missing scores are automatically excluded from the calculations. According to the competition rules, contestants with a total judges' score of zero are regarded as eliminated. Combined with official results, the elimination week for each contestant is extracted, thereby constructing a week-by-week elimination indicator.

On this basis, the proportion of each contestant's weekly judges' score relative to all remaining contestants is further computed, forming a judges' scoring metric that is directly comparable to the audience voting share. After these processing steps, a structurally consistent analytical dataset suitable for week-by-week elimination–constrained modeling is obtained.

3 Problem 1:Fan Vote Estimation

3.1 Solution Approach

Problem 1 is atypical inverse modeling problem. In the absence of publicly available audience voting results, only the weekly judges' scores and the actual elimination outcomes are known. The task is therefore to infer, in reverse, the proportion of audience votes received by each contestant in each week, such that the real elimination sequence can be reasonably reproduced under the officially announced scoring and elimination rules.

For this type of incomplete-information problem, this study does not aim to obtain a unique deterministic solution. Instead, it constructs a voting-estimation model that satisfies the relevant constraints, in order to characterize the feasible range of audience vote distributions and their associated uncertainty.

3.2 Model Formulation for Problem 1

The audience voting proportions satisfy the following basic constraints:

$$v_{s,i,t} \geq 0, \qquad \sum_{i \in A_{s,t}} v_{s,i,t} = 1, \quad \forall i \in A_{s,t}. \tag{1}$$

The problem involves two elimination mechanisms: percentage-based and rank-based combination. While both can be applied in the forward evaluation process, only the percentage-based rule provides a continuous and stable formulation suitable for inverse inference. Therefore, Problem 1 adopts the percentage-based mechanism to reconstruct audience voting proportions, and the two rules are compared in subsequent problems to evaluate their effects on elimination outcomes and fairness.

The judges' scores are normalized, and the judges' score proportion is defined as:

$$q_{s,i,t} = \frac{J_{s,i,t}}{\sum_{j \in A_{s,t}} J_{s,j,t}}, \qquad \sum_{i \in A_{s,t}} q_{s,i,t} = 1. \tag{2}$$

According to the problem description, the total percentage score is obtained by simple addition. Therefore, the overall score of contestant i in week t is defined as:

$$C_{s,i,t} = q_{s,i,t} + v_{s,i,t}, \qquad i \in A_{s,t}. \tag{3}$$

Given the observed elimination results, this implies that in week t , the eliminated contestant's overall score should rank among the lowest $k_{s,t}$positions among all contestants still competing in that week.

The above ranking condition can be transformed into the following system of inequality constraints:

$$C_{s,e,t} \leq C_{s,j,t} - \varepsilon, \qquad \forall e \in E_{s,t},\ \forall j \in A_{s,t} \setminus E_{s,t}, \tag{4}$$

The elimination week of each contestant is obtained by parsing the official results fields. For every season and each week, an elimination set is constructed based on the actual number of contestants eliminated in that week. If no elimination occurs in a given week, no elimination-related constraints are imposed for that week. This treatment ensures that the model constraints remain consistent with the actual competition format.

Because inverse problems typically admit multiple feasible solutions, relying solely on elimination constraints may lead to unrealistic or extreme voting structures. To obtain estimation results that better reflect actual audience behavior, this study introduces the following two rationality constraints.

(1)Maximum Entropy Principle The entropy function of the weekly audience voting distribution is defined as:

$$H_{s,t} = -\sum_{i \in A_{s,t}} v_{s,i,t} \log(v_{s,i,t}), \tag{5}$$

By maximizing the entropy function, excessive concentration of votes can be effectively avoided, resulting in more balanced and reasonable estimation outcomes.

(2)Temporal Smoothing Constraint Considering that audience preferences typically evolve smoothly over the course of the competition, penalties are imposed on large changes in voting proportions between adjacent weeks.

$$S_{s,t} = \sum_{i \in A_{s,t} \cap A_{s,t-1}} \left(v_{s,i,t} - v_{s,i,t-1}\right)^2. \tag{6}$$

A smoothing coefficient $\lambda \geq 0$ is introduced to control the strength of penalization on inter-week variations.

By integrating the above constraints and objective functions, the following optimization model is constructed for each season s :

$$\max_{\{v_{s,i,t}\}} \sum_t \left[-\sum_{i \in A_{s,t}} v_{s,i,t} \log(v_{s,i,t}) - \lambda \sum_{i \in A_{s,t} \cap A_{s,t-1}} \left(v_{s,i,t} - v_{s,i,t-1}\right)^2 \right] \tag{7}$$

The constraints are as follows:

$$\sum_{i \in A_{s,t}} v_{s,i,t} = 1, \qquad v_{s,i,t} \geq 0, \qquad \forall i \in A_{s,t},\ \forall t, \tag{8}$$

$$(q_{s,e,t} + v_{s,e,t}) \leq (q_{s,j,t} + v_{s,j,t}) - \varepsilon, \qquad \forall e \in E_{s,t},\ \forall j \in A_{s,t} \setminus E_{s,t}. \tag{9}$$

The solution $\hat{v}_{s,i,t}$obtained from the model represents the estimated audience voting proportions that, under the problem's simple additive rule, are able to reproduce the actual elimination path.

3.3 Model Solution for Problem 1

3.3.1 Properties and Solvability of the Optimization Problem

For a fixed season s, the model estimates the audience voting proportion vector $v_{s,\cdot,t} = \{v_{s,i,t}\}_{i \in A_{s,t}}$ for each competition week. The objective function consists of two components: a maximum entropy term and a temporal smoothing penalty.

The entropy term $-v \log v$ is concave over the domain $v \geq 0$, and the temporal smoothing term $-(v_{s,i,t} - v_{s,i,t-1})^2$ is also concave. The audience voting proportions satisfy the linear simplex constraints

$\sum_{i \in A_{s,t}} v_{s,i,t} = 1$ and $v_{s,i,t} \geq 0$ for all $i \in A_{s,t}$. Using the combined score definition $C_{s,i,t} = q_{s,i,t} + v_{s,i,t}$, the elimination consistency requirement is enforced by the linear inequality system $q_{s,e,t} + v_{s,e,t} \leq q_{s,j,t} + v_{s,j,t} - \varepsilon$ for all $e \in E_{s,t}$ and $j \in A_{s,t} \setminus E_{s,t}$. Therefore, the resulting optimization problem is a standard convex optimization problem.

As a result, for given parameters λ and ε, the model admits a globally optimal solution, ensuring numerical stability and reproducibility of the estimated voting proportions.

3.3.2 Parameter Selection and Feasibility Considerations

(1) Margin parameter ε The parameter ε is introduced solely to avoid degenerate ties in combined scores. Since both $q_{s,i,t}$ and $v_{s,i,t}$ are proportions, ε is chosen as a sufficiently small positive constant (In this study, the value $10^{-5}$is adopted.), ensuring that ranking relationships are preserved.

(2) Smoothing coefficient λ The coefficient $\lambda \geq 0$ controls the trade-off between faithfully reproducing the observed elimination path and maintaining temporal smoothness in voting behavior. Smaller values emphasize elimination consistency, while larger values impose stronger smoothness across weeks.

A sensitivity analysis over a range of candidate values (e.g., $\lambda \in \{0, 0.01, 0.1, 1, 10\}$) is conducted. The final parameter is selected to achieve both high elimination consistency and reasonable inter-week stability.

3.3.3 Model Output

The optimal solution $\hat{v}_{s,i,t}$ represents the estimated proportion of total audience votes received by contestant i in week t of season s, with weekly vote totals normalized to one.

By combining the estimated audience votes with normalized judge scores, the overall score

$$\hat{C}_{s,i,t} = q_{s,i,t} + \hat{v}_{s,i,t}$$

can be computed. These results serve as the primary inputs for subsequent analyses, including voting-system comparison, controversial contestant case studies, and fairness evaluation.

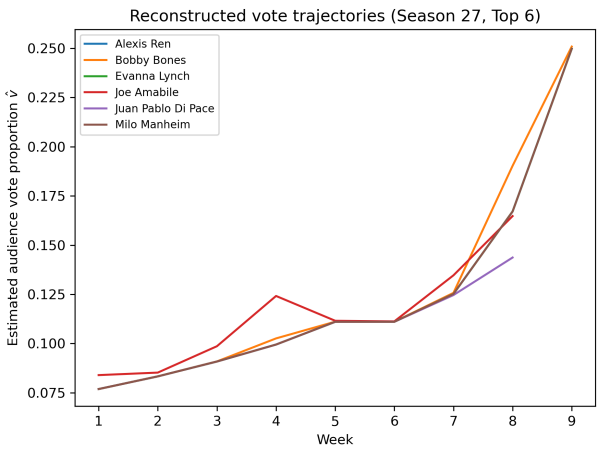
3.3.4 Reconstructed Voting Patterns and Judge–Audience Comparison

Figure 1a presents the reconstructed audience vote proportions $\hat{v}_{s,i,t}$ over time for Season 27, focusing on the six contestants with the highest average estimated support. The results show that the reconstructed vote shares evolve smoothly across weeks, without abrupt fluctuations between adjacent rounds. This behavior reflects the effect of the temporal smoothing term in the objective function and is consistent with the assumption that audience preferences change gradually as the competition progresses. Moreover, contestants with strong audience appeal exhibit a clear upward trend in vote proportions in later weeks, aligning with the increasing concentration of viewer attention toward finalists observed in practice.

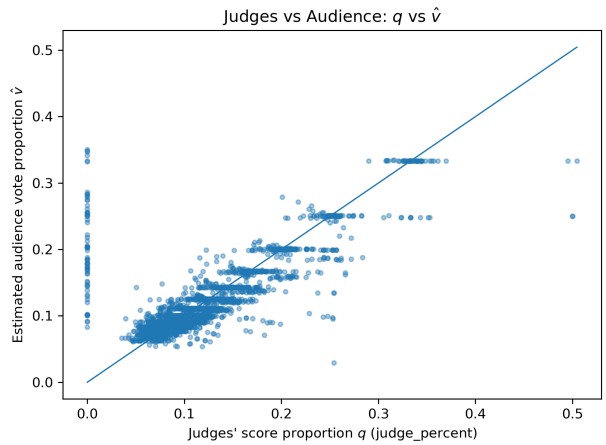
Figure 1b illustrates the relationship between judges' score proportions $q_{s,i,t}$ and the estimated audience vote proportions $\hat{v}_{s,i,t}$. The diagonal line represents the benchmark case where judges and the audience assign identical relative support. Most observations lie close to this diagonal, indicating

a high degree of overall consistency between judges' evaluations and audience preferences. However, a subset of points deviates noticeably from the diagonal, revealing instances where audience voting behavior differs substantially from judges' scores. These discrepancies provide direct evidence of potential judge–audience disagreement and motivate further analysis of controversial contestants and the fairness of the voting mechanism.

Overall, the visualized results demonstrate that the proposed inverse voting estimation model not only reproduces the observed elimination outcomes, but also yields interpretable and behaviorally plausible voting patterns, forming a solid foundation for subsequent comparisons of voting rules and fairness assessment.



(a) Reconstructed audience vote trajectories for the top six contestants in Season 27.



(b) Judges' score proportions versus estimated audience vote proportions in Season 27.

Figure 1: Visualization of reconstructed audience voting behavior and judge–audience comparison under the proposed inverse voting estimation model.

3.4 Consistency and Certainty Evaluation

To assess whether the reconstructed audience vote proportions are both *outcome-consistent* and *informative*, we introduce quantitative metrics evaluating elimination consistency and voting uncertainty based on the final model outputs.

3.4.1 Elimination Consistency

We first examine whether the estimated combined scores $C_{s,i,t} = q_{s,i,t} + \hat{v}_{s,i,t}$ correctly reproduce the observed elimination outcomes. For each elimination week, we check whether all eliminated contestants have combined scores no greater than those of any remaining contestants.

We define the *Elimination Consistency Rate (ECR)* as the fraction of elimination weeks in which this condition is satisfied. Across all 264 elimination weeks in the dataset, the model achieves

$$\text{ECR} = 1.00,$$

indicating that the reconstructed audience votes lead to elimination outcomes fully consistent with the official results in every case. This confirms that the inverse optimization model successfully enforces the elimination constraints and produces numerically feasible and stable solutions.

3.4.2 Elimination Margin and Decisiveness

While elimination consistency ensures correctness of ordering, it does not reflect how decisive each elimination is. To quantify the degree of separation between eliminated and surviving contestants, we define the *elimination margin*

$$\Delta_{s,t} = \min_{j \in A_{s,t} \setminus E_{s,t}} C_{s,j,t} - \max_{i \in E_{s,t}} C_{s,i,t},$$

which measures the smallest score gap separating eliminated contestants from those who remain.

Table 2 summarizes the distribution of elimination margins across all elimination weeks. The mean margin is approximately 0.0039, while the median margin is only 1×10^{-5}. This indicates that most eliminations are highly marginal rather than decisive.

Table 2: Summary statistics of elimination margins and vote entropy.

Metric	Count	Mean	Std	Min	Median	Max
Elimination margin Δ	264	0.0039	0.0097	1.0×10^{-5}	1×10^{-5}	0.082
Vote entropy H	355	2.055	0.418	1.099	2.164	2.773

Figure 2 shows the empirical distribution of elimination margins. The heavy concentration near zero visually confirms that most eliminations are determined by very small differences in combined scores.

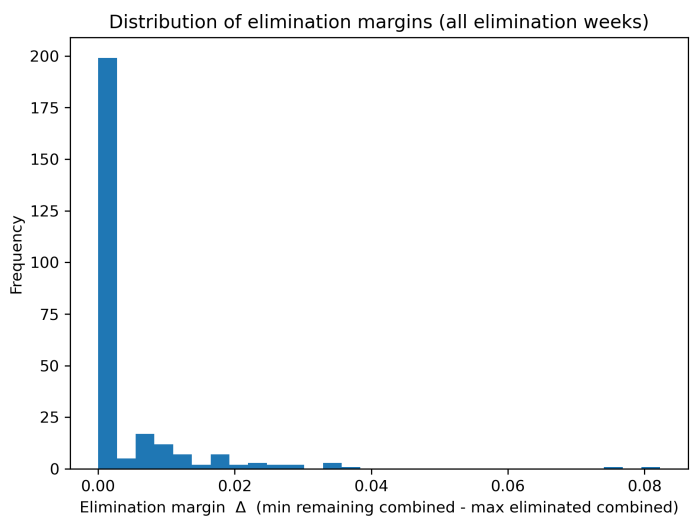


Figure 2: Distribution of elimination margins across all elimination weeks.

3.4.3 Voting Uncertainty via Entropy

To quantify uncertainty in reconstructed audience voting behavior, we compute the vote entropy for each competition week,

$$H_{s,t} = -\sum_{i \in A_{s,t}} \hat{v}_{s,i,t} \log \hat{v}_{s,i,t}.$$

Higher entropy indicates more dispersed and uncertain audience preferences, while lower entropy reflects concentrated support.

Across all weeks, the average entropy is 2.055, with values ranging from near zero in final rounds to approximately 2.773 in early stages with many active contestants. This pattern aligns with the structure of the competition: audience preferences are initially diffuse and gradually concentrate as the field narrows.

Figure 3 relates voting uncertainty to elimination decisiveness. While higher entropy generally corresponds to smaller elimination margins, substantial variability remains, suggesting that close eliminations can occur even under moderate uncertainty.

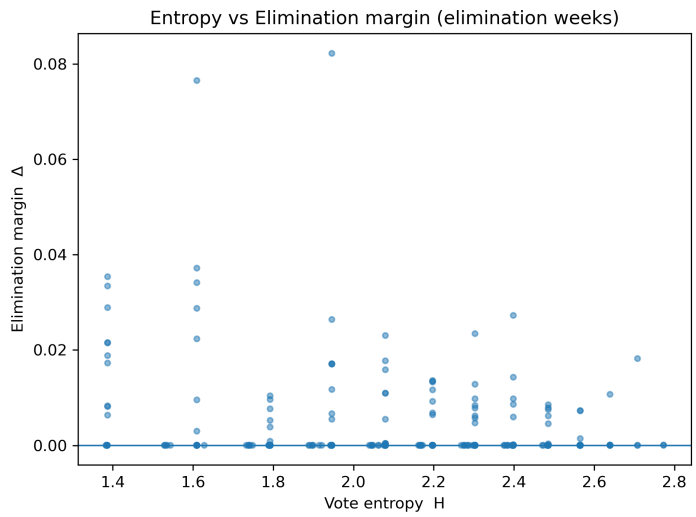
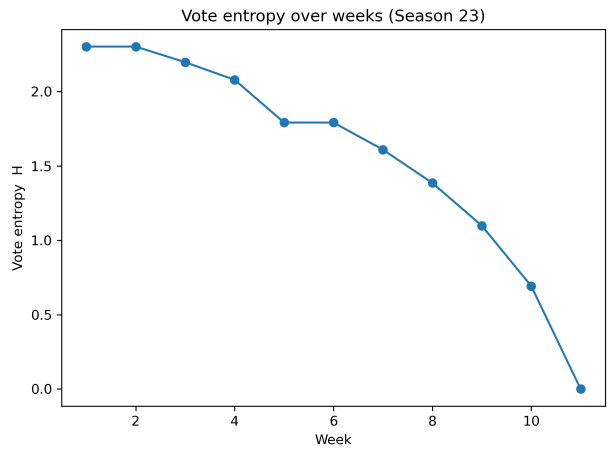


Figure 3: Vote entropy versus elimination margin across all elimination weeks. The horizontal reference line indicates the boundary $\Delta = 0$.

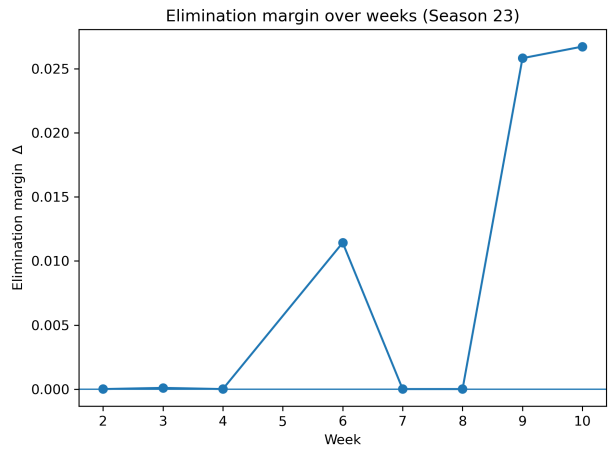
3.4.4 Season-Level Heterogeneity

Aggregating the metrics at the season level reveals systematic differences in elimination decisiveness. Several seasons exhibit consistently smaller average and minimum elimination margins.

Figures 4a and 4b illustrate the temporal evolution of vote entropy and elimination margins within a representative season. Entropy decreases steadily over time as contestants are eliminated, while elimination margins fluctuate and frequently approach zero, highlighting the fragile nature of weekly outcomes.



(a) Vote entropy over weeks (Season 23).



(b) Elimination margin over weeks (Season 23).

Figure 4: Within-season dynamics of voting uncertainty and elimination decisiveness.

3.4.5 Discussion

Together, these results demonstrate that the proposed inverse voting estimation model not only reproduces elimination outcomes exactly, but also captures the intrinsic uncertainty and marginal nature of audience-driven voting. Eliminations are often determined by narrow score differences, and the degree of certainty varies substantially across weeks and seasons. These properties provide a rigorous quantitative foundation for subsequent comparisons of voting rules and assessments of fairness in competitive elimination formats.

4 Problem 2: Voting Rule Comparison and Fairness Analysis

4.1 Solution Approach

Problem 2 focuses on the systematic impact of voting combination rules themselves on competition outcomes. In Problem 1, audience votes have been estimated as a set of reasonable inputs under the given information constraints. Building on this foundation, the present problem treats the estimated audience votes as fixed and varies only the method used to combine judges' scores and audience votes. By comparing the resulting elimination paths and final rankings under different institutional arrangements, the analysis evaluates the extent to which voting rules modulate competition outcomes.

This problem does not seek to determine the "true" or "correct" outcome for any specific season or contestant. Instead, through cross-season and cross-rule comparative analysis, it addresses the following core questions: Do different voting combination rules systematically favor audience preferences or judges' evaluations? Under what conditions are such biases amplified or mitigated?

4.2 Formal Definition of Voting Rules

Let s denote the season and t the week. Let $A_{s,t}$ be the set of contestants still active in week t of season s. For contestant $i \in A_{s,t}$, denote the estimated fan vote share by $\hat{v}_{s,i,t}$ and the total judges' score

by $J_{s,i,t}$. Let $k_{s,t}$ be the number of contestants eliminated in week t.

Rule 1: Percentage-Based Combination (Percent) The judges' score share is defined as

$$q_{s,i,t} = \frac{J_{s,i,t}}{\sum_{j \in A_{s,t}} J_{s,j,t}}.$$

The percentage-based combined score is given by

$$C^{(P)}_{s,i,t} = q_{s,i,t} + \hat{v}_{s,i,t}.$$

The elimination set is then defined as

$$E^{(P)}_{s,t} = \text{Bottom-}k_{s,t}\big(C^{(P)}_{s,i,t} : \ i \in A_{s,t}\big),$$

i.e., the $k_{s,t}$ contestants with the lowest combined scores are eliminated.

Rule 2: Rank-Based Combination (Rank) Contestants are ranked separately according to judges' scores and fan votes. Define the judges' rank and fan vote rank as

$$r^{(J)}_{s,i,t} = \text{RankAsc}(-J_{s,i,t}), \qquad r^{(V)}_{s,i,t} = \text{RankAsc}(-\hat{v}_{s,i,t}),$$

where $\text{RankAsc}(\cdot)$ assigns ranks $1, 2, \ldots$ in ascending order. The rank-based combined score is defined as

$$C^{(R)}_{s,i,t} = r^{(J)}_{s,i,t} + r^{(V)}_{s,i,t}.$$

The elimination set is determined by

$$E^{(R)}_{s,t} = \text{Top-}k_{s,t}\big(C^{(R)}_{s,i,t} : \ i \in A_{s,t}\big),$$

i.e., the $k_{s,t}$ contestants with the largest rank sums are eliminated.

Rule 3: Bottom-Two with Judges' Save (Bottom-2 + Judges) This mechanism consists of two stages. Let the base combination rule be $\mathcal{M} \in \{P, R\}$.

Stage 1: Compute the combined scores $C^{(\mathcal{M})}_{s,i,t}$ and identify the bottom two contestants:

$$B^{(\mathcal{M})}_{s,t} = \text{Bottom-2}\big(C^{(\mathcal{M})}_{s,i,t} : \ i \in A_{s,t}\big).$$

Stage 2: Judges select the contestant with the lower judges' score from the bottom two for elimination:

$$e^{(\mathcal{M}+J)}_{s,t} = \arg\min_{i \in B^{(\mathcal{M})}_{s,t}} J_{s,i,t}.$$

If $k_{s,t} = 1$, the elimination set is

$$E^{(\mathcal{M}+J)}_{s,t} = \{e^{(\mathcal{M}+J)}_{s,t}\}.$$

If $k_{s,t} > 1$, the two-stage procedure is applied iteratively until $k_{s,t}$ contestants are eliminated.

4.3 Deviations from the Percentage-Based Rule under Alternative Voting Mechanisms

Using the estimated fan vote shares obtained in Problem 1, the percentage-based aggregation rule is able to perfectly reproduce the observed historical elimination outcomes in Task 2. Under this empirical result, the purpose of this section is no longer to evaluate the relative accuracy of different voting rules in a symmetric manner. Instead, we treat the percentage-based rule as a benchmark mechanism and examine how alternative voting rules deviate from this benchmark, as well as under what conditions such deviations are most likely to occur.

Frequency of deviation under the Rank-based rule We first compare the Rank-based rule to the percentage-based benchmark by identifying elimination weeks in which the two rules eliminate different contestants. For each season, we compute the proportion of elimination weeks in which the Rank rule deviates from the benchmark outcome. The distribution of this deviation frequency across seasons is shown in Figure 5.

The results indicate that deviations are relatively infrequent in most seasons, with the Rank rule coinciding with the benchmark in the majority of elimination weeks. However, several seasons exhibit non-negligible deviation rates, demonstrating that compressing vote information from continuous magnitudes to ordinal rankings can alter elimination decisions in specific cases, even when the benchmark rule fully explains historical outcomes.

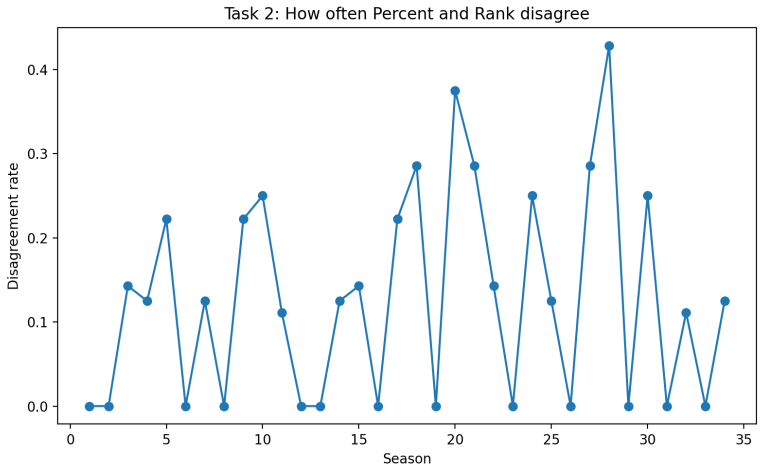


Figure 5: Season-level frequency of deviations between the Rank rule and the percentage-based benchmark

Stage-dependent deviation patterns To further characterize the conditions under which deviations arise, we group elimination weeks by the number of active contestants and compute the average deviation rate between the Rank rule and the percentage-based benchmark within each competition stage. The results are presented in Figure 6.

Deviations are more likely to occur in the early and middle stages of the competition, when multiple contestants cluster near the elimination boundary and score differences are relatively small. As the

competition progresses and fewer contestants remain, the Rank rule increasingly converges to the benchmark outcome. This pattern suggests that alternative aggregation rules primarily affect elimination decisions in borderline situations, rather than systematically reshaping the entire competition.

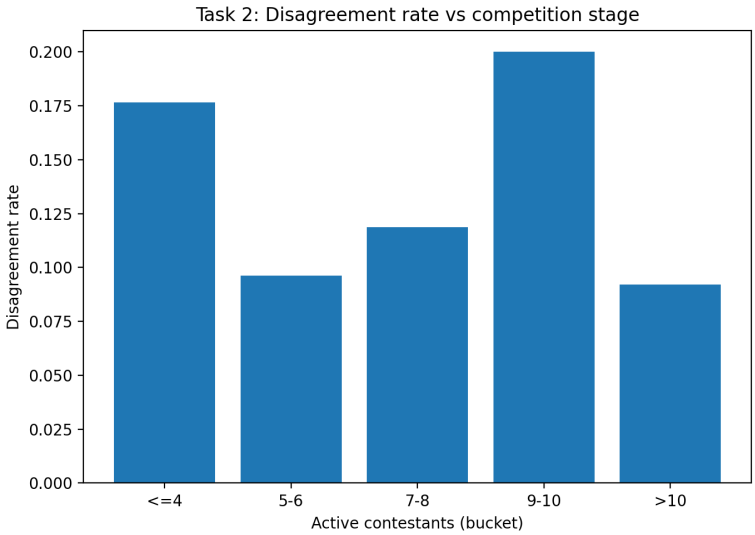


Figure 6: Average deviation rate between the Rank rule and the percentage-based benchmark across competition stages

Systematic effects of introducing judges' discretion We next examine the impact of introducing an additional decision stage through the Bottom-Two with Judges' Save mechanism. Figure 7 compares the elimination outcomes generated by the percentage-based rule, the Rank rule, and the Bottom-Two with Judges' Save mechanism against the observed historical eliminations.

Unlike the Rank rule, which deviates only in a limited subset of weeks, the Bottom-Two with Judges' Save mechanism produces systematic and persistent deviations from the percentage-based benchmark across seasons. This result indicates that reallocating decision power at the elimination boundary from aggregated audience preference to judges' discretion fundamentally alters elimination paths, even when underlying performance and voting preferences remain unchanged.

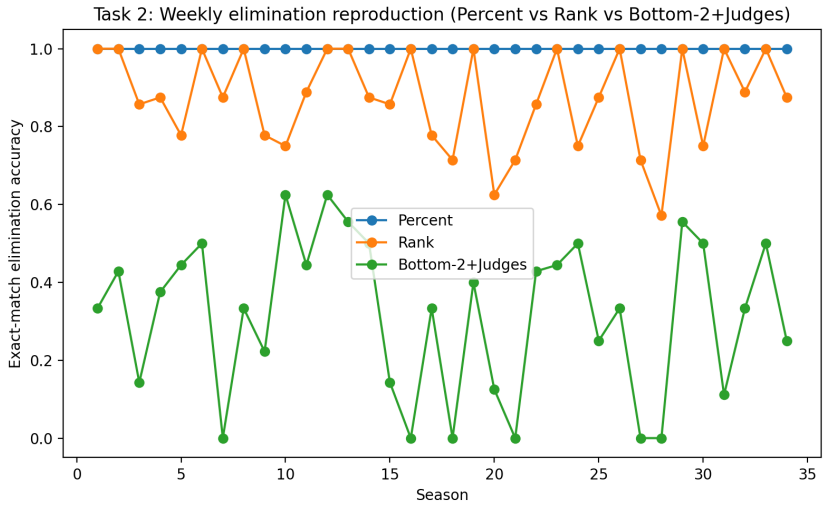


Figure 7: Elimination outcomes under alternative voting rules relative to the percentage-based benchmark

Summary In summary, when treated as a benchmark mechanism, the percentage-based rule fully explains observed elimination outcomes. The Rank-based rule remains largely consistent with this benchmark, but introduces limited deviations concentrated in early and mid-season stages. By contrast, the introduction of judges' discretion through a two-stage elimination mechanism leads to substantially larger and more systematic departures from benchmark outcomes. These aggregate deviation patterns provide a clear reference point for the contestant-level analyses of controversial cases in the following section.

4.4 Case Study

To further investigate how voting rule design influences elimination outcomes when judges' evaluations and fan preferences diverge, we conduct a focused case study on four controversial contestants: Jerry Rice (Season 2), Billy Ray Cyrus (Season 4), Bristol Palin (Season 11), and Bobby Bones (Season 27). These contestants are widely regarded as controversial because they repeatedly received low judges' scores while maintaining strong fan support, allowing them to survive far longer than their technical evaluations alone would suggest.

Figures 8–11 illustrate the weekly score composition of each contestant using clustered stacked bar charts. In each figure, the lower segment of each bar represents judges' score share, while the upper segment represents fan vote share estimated in Problem 1. Arrows mark the predicted elimination rounds under different voting rules, allowing a direct comparison of outcomes across aggregation methods.

Season 2: Jerry Rice. As shown in Figure 8, Jerry Rice consistently exhibits a score structure characterized by low judges' scores and relatively strong fan support. Under both the percentage-based rule and the rank-based rule, this fan support is sufficient to prevent early elimination, leading to identical predicted outcomes that align with the historical result in which he reached the final. However, once the Bottom-2 + Judges Save rule is introduced, Jerry Rice is eliminated in the later stage

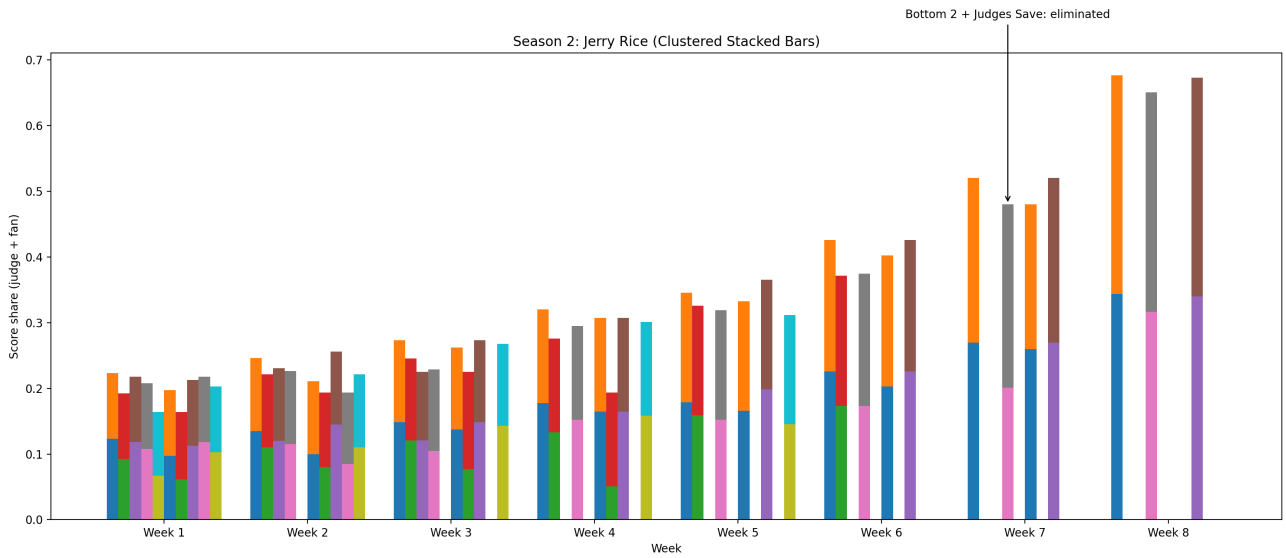


Figure 8: Season 2: Jerry Rice. Clustered stacked bar chart showing weekly score composition. In each bar, the lower segment represents judges' score share and the upper segment represents fan vote share estimated in Problem 1. Arrows indicate the predicted elimination round under different voting rules.

of the competition. This change highlights the key role of judges' final decision power: even when fan support is substantial, granting judges direct authority at the elimination stage can overturn outcomes that would otherwise persist under purely aggregated rules.

Season 4: Billy Ray Cyrus. Figure 9 shows that for Billy Ray Cyrus, the percentage-based and rank-based rules again yield similar elimination rounds, suggesting that the conflict between judges and fans is relatively moderate in this season. Nevertheless, under the Bottom-2 + Judges Save mechanism, his elimination occurs noticeably earlier. This indicates that when a contestant repeatedly appears near the bottom in judges' evaluations, the judges' save mechanism does not necessarily act as a corrective tool; instead, it may accelerate elimination by reinforcing judges' technical preferences.

Season 11: Bristol Palin. Bristol Palin represents one of the clearest examples of judge–fan preference divergence. As illustrated in Figure 10, both the percentage-based and rank-based rules allow her to survive deep into the competition, consistent with her actual third-place finish. However, the Bottom-2 + Judges Save rule dramatically alters this outcome, resulting in elimination in the middle rounds. This case demonstrates that the judges' save mechanism can substantially disadvantage contestants whose success is primarily driven by fan support rather than technical scores.

Season 27: Bobby Bones. The case of Bobby Bones, shown in Figure 11, exhibits the strongest sensitivity to voting rule design. Under the percentage-based rule, he survives throughout the season, matching his historical championship result. Under the rank-based rule, he is eliminated much earlier due to consistently low judges' rankings. Introducing the Bottom-2 + Judges Save mechanism partially moderates this effect, but his elimination risk remains significantly higher than under the percentage-

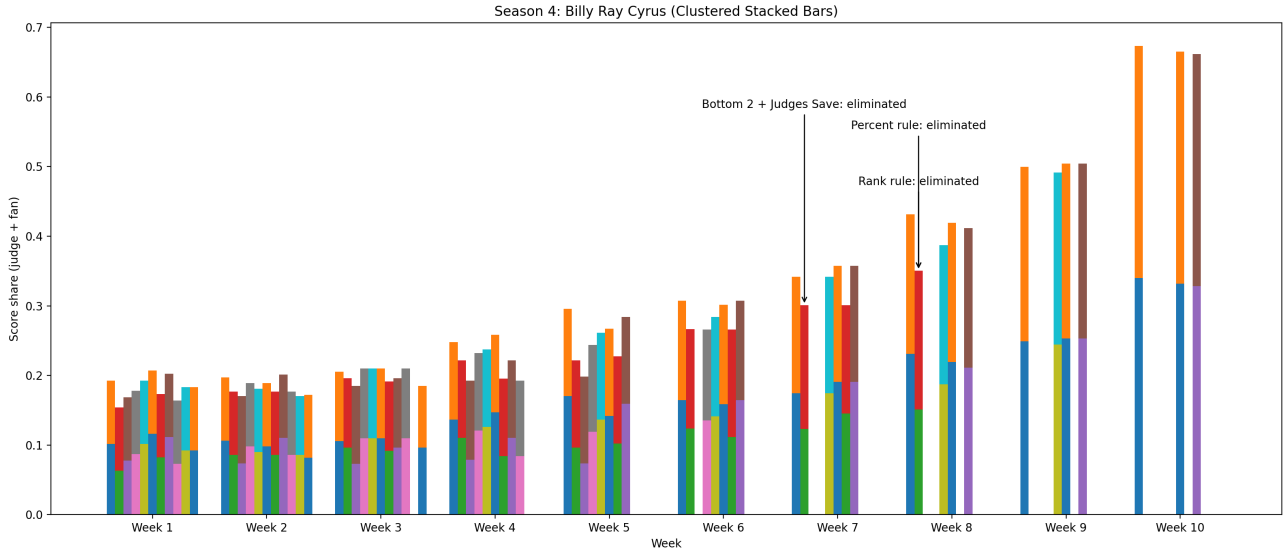


Figure 9: Season 4: Billy Ray Cyrus. Weekly clustered stacked bar chart illustrating the combined effects of judges' scores and fan votes. Arrows mark the elimination rounds under the percentage-based rule, rank-based rule, and Bottom 2 + Judges Save rule.

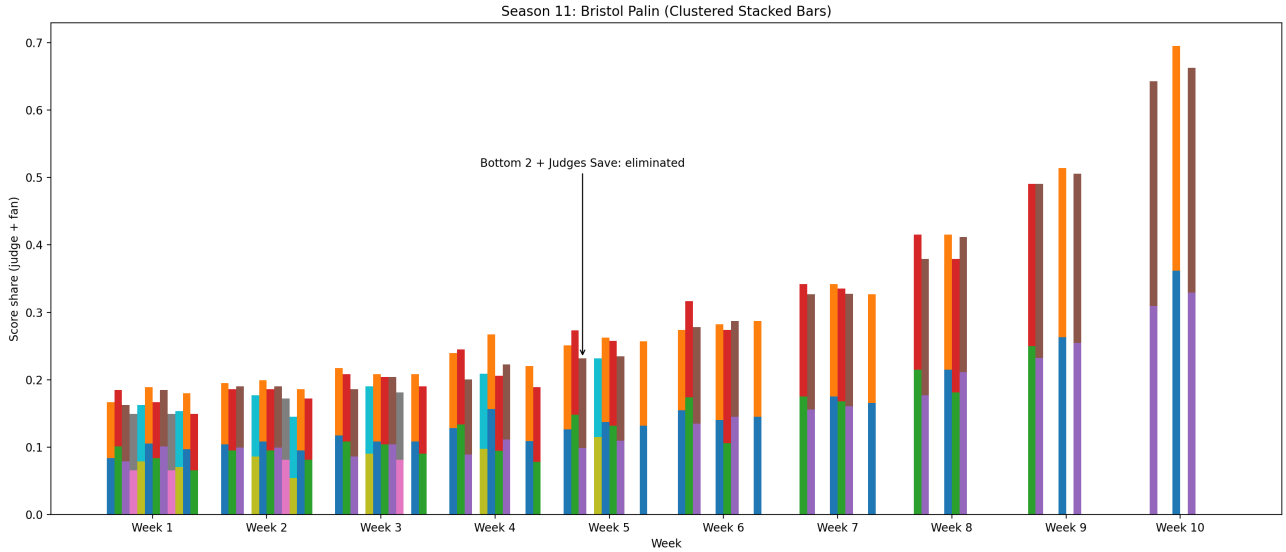


Figure 10: Season 11: Bristol Palin. Clustered stacked bars show the divergence between judges' evaluations and fan support across weeks. The figure highlights how different voting rules lead to different elimination outcomes for a fan-supported contestant.

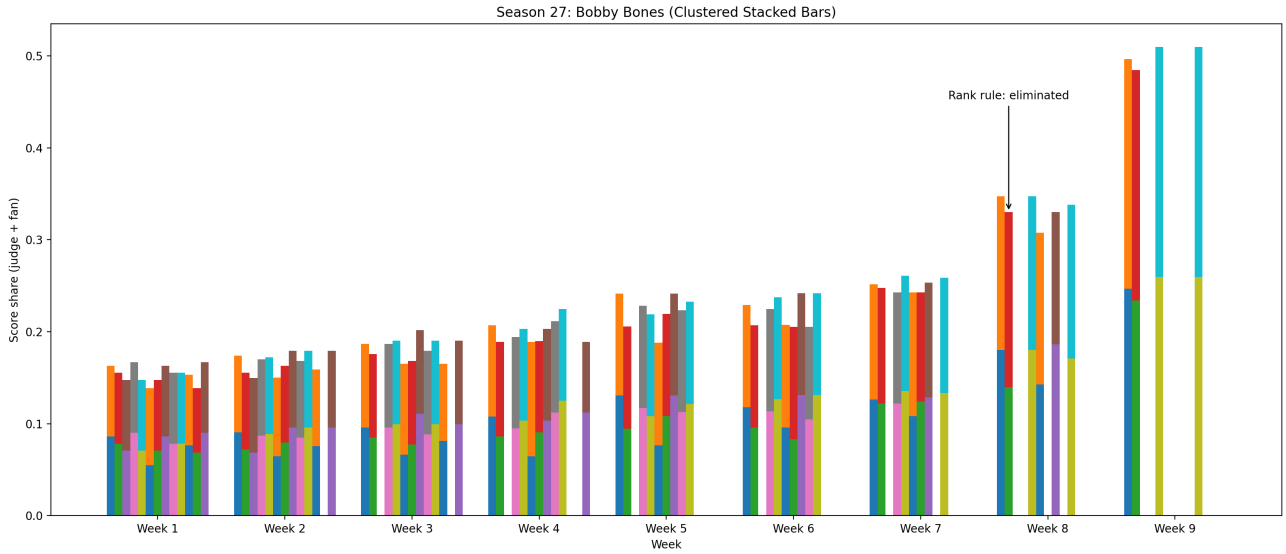


Figure 11: Season 27: Bobby Bones. Weekly score composition under judges' and fans' voting. The figure illustrates the strong dependence of elimination outcomes on the aggregation rule when judges' rankings are consistently low.

based rule, illustrating that judges' intervention can only partially offset the structural disadvantages imposed by rank aggregation.

Overall Discussion. Across all four cases, the Bottom-2 + Judges Save rule consistently shifts elimination outcomes toward judges' technical evaluations. While this mechanism is often intended to reduce controversial eliminations, our analysis shows that it can also introduce new forms of bias by systematically disadvantaging fan-supported contestants. Compared with the percentage-based rule, which generally produces outcomes closest to historical results, the Bottom-2 + Judges Save mechanism reallocates decision power at critical stages of the competition, thereby playing a decisive role in determining survival outcomes beyond contestants' overall score composition.

4.5 Comparison of Voting Rules and the Role of Judges' Intervention

Based on the modeling results and the case studies of controversial contestants presented above, we further evaluate the relative merits of different voting aggregation rules and discuss whether the introduction of judges' intervention is justified in determining elimination outcomes.

From an information perspective, the percentage-based rule is superior to the rank-based rule. By directly aggregating judges' score shares and fan vote shares, the percentage-based rule preserves the magnitude of support from both sources and reflects differences in voting intensity. In contrast, the rank-based rule retains only relative ordering information and discards absolute differences in scores and votes. While this loss of information may be inconsequential when judges' and fans' preferences align, it becomes critical when the two diverge, as small differences in judges' rankings can disproportionately affect elimination outcomes.

Empirically, the percentage-based rule produces outcomes that are more consistent with historical results. Across the four representative controversial cases examined in this study, the percentage-based

rule generally replicates the observed survival trajectories, whereas the rank-based rule frequently predicts substantially earlier eliminations. This suggests that the percentage-based rule more effectively captures the intended balance between professional evaluation and audience participation.

Moreover, the rank-based rule tends to amplify judges' preferences when systematic discrepancies exist between judges' evaluations and fan support. Because it is insensitive to the absolute scale of fan votes, contestants with strong popular backing but persistently low judges' rankings face a heightened risk of early elimination. This structural feature leads to biased outcomes for fan-supported contestants and increases the sensitivity of results to judges' subjective assessments.

Regarding the Bottom-2 + Judges Save mechanism, its original purpose is to prevent technically strong contestants from being eliminated prematurely due to insufficient fan support. However, the case studies demonstrate that when judges' and fans' preferences diverge sharply, this mechanism often fails to serve as a balancing tool. Instead, by granting judges final authority at the elimination stage, it systematically shifts decision power toward technical evaluations and disadvantages contestants whose success is primarily driven by fan votes.

Therefore, we argue that judges' intervention should not be adopted as a routine final decision rule. A more appropriate design would restrict such intervention to exceptional circumstances, such as clear scoring anomalies or voting irregularities, where corrective action is warranted. Overall, the percentage-based rule provides a more stable and equitable aggregation framework, while judges' intervention should function as a limited safeguard rather than a central component of the elimination process.

5 Problem 3: Effects of Professional Dancers and Celebrity Characteristics

5.1 Solution Approach

In the previous analysis, this study has estimated the audience voting proportions received by each contestant in each week. Building on these results, Task 3 further examines the impact of professional dancers and individual celebrity characteristics on competition performance, and compares whether these factors exert consistent effects on judges' scores and audience votes.

Specifically, the proportion of judges' scores and the proportion of audience votes are treated as separate outcome variables. Within a unified analytical framework, explanatory variables such as professional dancer status and celebrity characteristics—including age and industry background—are introduced to model the two outcomes separately. Week-specific effects are controlled for in order to eliminate differences in overall scoring and voting environments across different competition weeks.

By comparing the estimated effects of the same factors in the judges' score model and the audience vote model, this study reveals the differing mechanisms through which professional dancers and celebrity characteristics influence technical evaluations and audience preferences.

5.2 Model Formulation for Problem 3

To identify the effects of celebrity age and industry on competition outcomes and to assess the robustness of these effects, we adopt a stepwise regression strategy and construct Models $M1$–$M4$.

All models share the same dependent variable but differ in the set of explanatory and control variables included, which ensures comparability across specifications.

Model $M1$ serves as the baseline specification and is designed to control for the most prominent sources of structural heterogeneity in the competition. Specifically, $M1$ includes professional dancer fixed effects $\alpha_{\text{dancer}(i)}$, week fixed effects γ_t, and season fixed effects δ_s, and is given by

$$Y_{i,t} = \alpha_{\text{dancer}(i)} + \gamma_t + \delta_s + \varepsilon_{i,t}.$$

Professional dancers, weekly scoring scales, and season-specific conditions have substantial impacts on observed outcomes, yet they are not the primary variables of interest in this study. Controlling for these factors in the baseline model provides a clean reference point for evaluating the marginal contributions of celebrity characteristics in subsequent models.

Building on $M1$, Model $M2$ introduces celebrity age Age_i to examine whether age has an independent effect on competition outcomes after accounting for dancer and competition environment effects:

$$Y_{i,t} = \alpha_{\text{dancer}(i)} + \beta_1 \text{Age}_i + \gamma_t + \delta_s + \varepsilon_{i,t}.$$

Model $M3$ further extends $M2$ by incorporating celebrity industry Industry_i, allowing us to assess the marginal impact of professional background and to evaluate whether the inclusion of industry alters the estimated age effect:

$$Y_{i,t} = \alpha_{\text{dancer}(i)} + \beta_1 \text{Age}_i + \beta_2 \text{Industry}_i + \gamma_t + \delta_s + \varepsilon_{i,t}.$$

Finally, Model $M4$ augments the specification with additional background controls $\mathbf{Z}_i$, such as regional characteristics of the celebrity, yielding the full model:

$$Y_{i,t} = \alpha_{\text{dancer}(i)} + \beta_1 \text{Age}_i + \beta_2 \text{Industry}_i + \boldsymbol{\theta}^\top \mathbf{Z}_i + \gamma_t + \delta_s + \varepsilon_{i,t}.$$

By comparing coefficient estimates across Models $M1$–$M4$, we can systematically evaluate how the effects of age and industry evolve as additional controls are introduced, and assess the stability and robustness of the main results.

5.3 Model Solution for Problem 3

Table 3: Stepwise Regression Results: Judges' Scores

	M1	M2	M3	M4
Age		-0.00062^{***}	-0.00066^{***}	-0.00080^{***}
		(0.00014)	(0.00017)	(0.00018)
Industry Dummies			Yes	Yes
Professional Dancer FE	Yes	Yes	Yes	Yes
Week FE	Yes	Yes	Yes	Yes
Season FE	Yes	Yes	Yes	Yes
Regional Controls	No	No	No	Yes
R^2	0.660	0.671	0.679	0.696
Observations	2470	2470	2470	2470

Notes: The dependent variable is judges' score percentage. Standard errors clustered at the celebrity level are reported in parentheses. Industry is included as a set of dummy variables. *** denotes significance at the 1% level.

Table 4: Stepwise Regression Results: Estimated Fan Votes

	M1	M2	M3	M4
Age		0.00014^{*}	0.00016^{*}	0.00020^{**}
		(0.00007)	(0.00009)	(0.00009)
Industry Dummies			Yes	Yes
Professional Dancer FE	Yes	Yes	Yes	Yes
Week FE	Yes	Yes	Yes	Yes
Season FE	Yes	Yes	Yes	Yes
Regional Controls	No	No	No	Yes
R^2	0.800	0.801	0.804	0.807
Observations	2470	2470	2470	2470

Notes: The dependent variable is the estimated fan vote share. Standard errors clustered at the celebrity level are reported in parentheses. ** and * denote significance at the 5% and 10% levels, respectively.

The stepwise regression results provide clear evidence on how celebrity characteristics affect judges' evaluations and fan voting behavior. In the baseline specification $M1$, variation in outcomes is primarily explained by professional dancer effects and competition-level factors, indicating that partnership quality and the competitive environment account for a substantial share of observed differences. When celebrity age is introduced in $M2$, the estimated coefficient on Age_i is statistically significant in the judges' score model, while its effect on estimated fan vote shares is comparatively weak. This suggests that judges are more responsive to age-related factors, such as physical capability or technical consistency, than the viewing audience.

In $M3$, the inclusion of celebrity industry leads to a marked improvement in the explanatory power of the fan vote model, whereas its contribution to the judges' score model remains limited. This pattern indicates that professional background plays a more prominent role in shaping audience preferences than technical evaluations by judges. Importantly, the estimated age effect remains stable after industry is added, implying that the two characteristics capture distinct dimensions of celebrity influence. Finally, in the full specification $M4$, which controls for additional background characteristics, the signs and magnitudes of the age and industry coefficients remain largely unchanged across both outcome variables. The consistency of these estimates across models confirms the robustness of the main findings.

6 Problem 4: An Improved Elimination System

6.1 Rule Design

The previous analysis demonstrates that judges' scores and fan votes capture different dimensions of contestant performance. Judges tend to emphasize technical execution and consistency, while fan voting reflects popularity and public appeal. Relying exclusively on either component may therefore lead to controversial eliminations. To address this issue, we propose an alternative elimination system that integrates judges' scores and fan votes in a way that improves fairness while maintaining audience engagement.

Specifically, in each competition week, we define a continuous composite score based on judges' score percentages and fan vote percentages. For contestant i in week t, the composite score is given by

$$S_{i,t} = \lambda_t J_{i,t} + (1 - \lambda_t) V_{i,t},$$

where $J_{i,t}$ denotes the judges' score percentage and $V_{i,t}$ denotes the estimated fan vote share. The weight λ_t is allowed to vary over the course of the season. Early in the competition, a relatively lower value of λ_t assigns greater importance to fan votes, encouraging audience participation and discussion. As the season progresses, λ_t gradually increases, placing more emphasis on judges' evaluations to ensure that finalists and the eventual winner exhibit strong technical performance. This dynamic weighting balances entertainment value with competitive credibility.

To further prevent technically strong contestants from being eliminated due to short-term fluctuations in popularity, we introduce a merit safeguard mechanism. If a contestant's judges' score in a given week ranks within the top fraction of all contestants (for example, the top 70%), that contestant is protected from direct elimination in that week. Even if their composite score falls into the elimination range, the elimination candidate is instead selected from contestants who do not meet the safeguard criterion and have lower composite scores. This safeguard establishes a minimum technical threshold while preserving the decisive role of fan votes in close elimination scenarios.

The proposed system achieves a balance across fairness, stability, and viewer experience. From a fairness perspective, the merit safeguard substantially reduces the likelihood that contestants highly rated by judges are eliminated prematurely. From the audience perspective, fan votes continue to exert meaningful influence on weekly outcomes, ensuring that viewer participation remains consequential. From a methodological standpoint, the composite score is a continuous function, making the elimination outcome less sensitive to small perturbations in votes than rank-based rules, while remaining transparent and easy to explain.

6.2 Evaluation

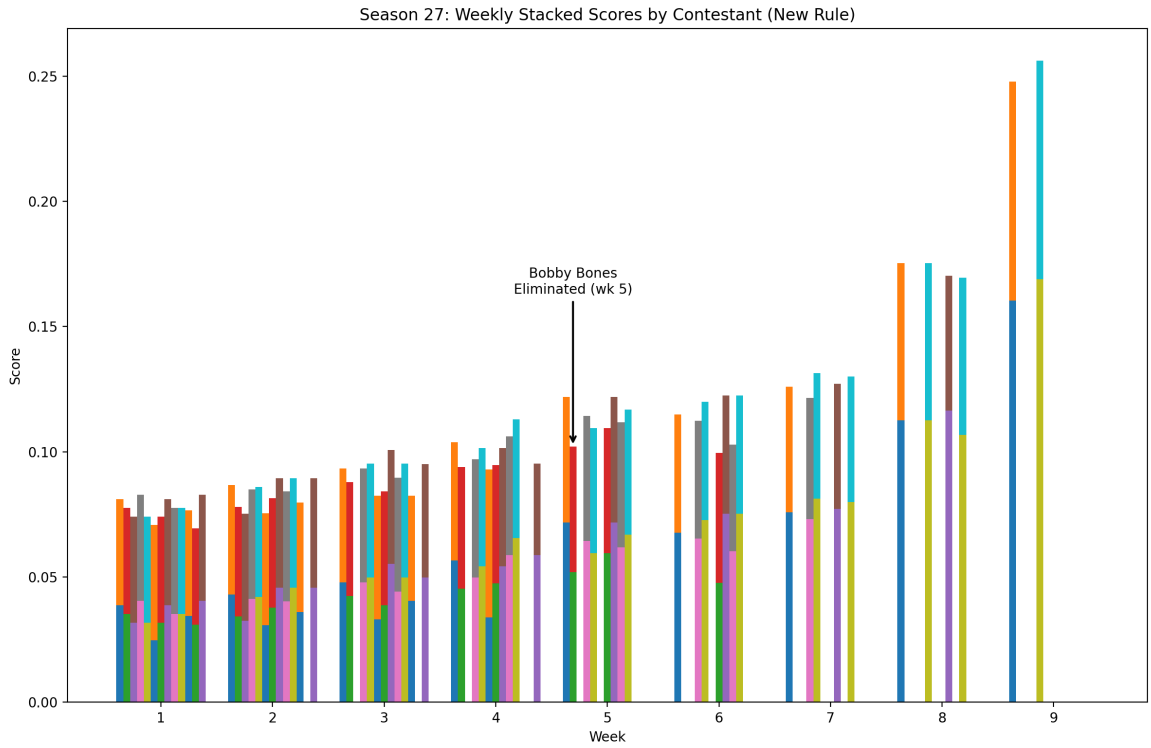


Figure 12: Season 27: Weekly stacked scores under the proposed new voting rule. The stacked bars represent the dynamically weighted combination of judges' scores and fan votes. The arrow indicates the predicted elimination round of Bobby Bones under the new rule.

6.3 Effectiveness of the Proposed Voting Rule: Evidence from Season 27

To illustrate the effectiveness of the proposed voting rule, we examine Season 27 and focus on the contestant Bobby Bones as a representative case. Historically, Bobby Bones received consistently low judges' scores throughout the season, yet ultimately won the competition due to exceptionally strong fan support. This season therefore provides an ideal test case for evaluating whether a voting rule can balance professional evaluation and audience preference without relying on direct judges' intervention.

Figure 12 presents the weekly score composition of contestants in Season 27 under the proposed rule. Each stacked bar represents the dynamically weighted combination of judges' score shares and fan vote shares, as defined in the new aggregation framework. The arrow marks the round in which Bobby Bones is predicted to be eliminated under the proposed rule.

Under the new rule, Bobby Bones is eliminated in Week 5. Unlike the rank-based rule, this outcome is not driven by an excessive amplification of judges' relative rankings, but instead reflects his persistently low judges' scores combined with the gradual increase in judges' weight over time. At the same time, the elimination does not result from a discrete judges' decision at the elimination stage, as in the Bottom-2 + Judges Save mechanism. Rather, the proposed rule allows judges' influence to

increase smoothly through dynamic weighting and the technical protection band, avoiding abrupt shifts in decision authority.

This result highlights several advantages of the proposed rule. First, it prevents the sudden concentration of judges' power at a single elimination round, ensuring that elimination risk evolves continuously as the competition progresses. Second, by incorporating a technical protection band, the rule distinguishes between contestants with consistently strong technical performance and those relying primarily on fan support, reducing the likelihood that technically strong contestants are eliminated due to short-term fluctuations in fan votes. Third, the proposed rule preserves the influence of fan voting while preventing it from dominating the outcome when judges' evaluations remain persistently low.

Overall, the Season 27 case demonstrates that the proposed voting rule offers a more transparent, balanced, and controllable alternative to both the rank-based rule and the Bottom-2 + Judges Save mechanism. By integrating judges' evaluations through structural design rather than direct intervention, the new rule achieves a smoother trade-off between fairness and professionalism when judges' and fans' preferences diverge.